

Étale Cohomology of Curves

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1 Introduction

The goal of this paper is to give a short proof of the following.

Theorem 1.1. Let k be a separably closed field and let X/k be a connected complete algebraic curve. Let n be an integer relative prime to the characteristic of k . Then the étale cohomology of the sheaf of primitive n -th roots of unit on X can be computed as

$$H_{\text{ét}}^*(X, (\mu_n)_X) = \begin{cases} \mu_n(X) & \text{for } * = 0, \\ {}_n\text{Pic}(X) & \text{for } * = 1, \\ \text{Pic}_n(X) & \text{for } * = 2, \end{cases}$$

for lower degrees. In higher degrees it vanishes. Here ${}_n\text{Pic}(X)$ and $\text{Pic}_n(X)$ denote the kernel and cokernel of the multiplication by n on $\text{Pic}(X)$. Hence ${}_n\text{Pic}(X)$ can be identified with the n -torsion subgroup of $\text{Pic}(X)$.

Remark 1.2. By choosing a n -th primitive root of unit, we obtain an isomorphism $(\mu_n)_X \cong (\mathbb{Z}/n)_X$. In this situation, [Theorem 1.1](#) computes the cohomology of the constant sheaf $(\mathbb{Z}/n)_X$, though non canonically.

Remark 1.3. Let k be a separably closed field and let X/k be a connected complete algebraic curve. Assume also that X is smooth. Then the kernel of the multiplication by n on $\text{Pic}(X)$ is actually given by ${}_n\text{Pic}(X) \cong (\mathbb{Z}/n)^{2g}$, where g is the genus of X . By choosing a n -th primitive root of unit, we obtain isomorphisms $(\mu_n)_X \cong (\mathbb{Z}/n)_X$ and $\text{Pic}(X)_n \cong \mathbb{Z}/n$. In particular, the cohomology groups of $(\mu_n)_X \cong (\mathbb{Z}/n)_X$ have rank 1, $2g$ and 1 as $\mathbb{Z}/n$ -modules.

Notation 1.4. In the following pages we will make our life easier with some *ad hoc* notation. Let X be a scheme. We will denote by $\text{cl}(X)$ the set of closed points of X . By $X_{\text{ét}}$ we will denote the étale site of X and by $\text{Shv}_{\text{Ab}}(X_{\text{ét}})$ we will denote the 1-category of abelian sheaves on the étale site of X . Given a point $x \in X$ we will denote by $\bar{x}$ the associated geometric point, that is the canonical morphism $\bar{x} : \text{Spec}(k(x)^{\text{sep}}) \rightarrow X$ for a choice of the separable closure $k(x)^{\text{sep}}$ of the residue field $k(x)$. Let $\mathcal{F} \in \text{Shv}_{\text{Ab}}(X_{\text{ét}})$ be an abelian sheaf on the étale site of X . We will denote by $\mathcal{F}_{\bar{x}} = \bar{x}^*(\mathcal{F})$ the stalk at the geometric point $\bar{x}$.

2 Skyscraper sheaves

Definition 2.1. Let X be a scheme and let $\mathcal{F} \in \text{Shv}_{\text{Ab}}(X_{\text{ét}})$ be an abelian sheaf on the étale site of X . We will say that $\mathcal{F}$ is a *skyscraper sheaf* if the stalk $\mathcal{F}_{\bar{x}} = 0$ vanishes for all non-closed points $\bar{x}$ of X .

In order to efficiently use this definition we need an equivalent characterization.

Lemma 2.2. Let X be a noetherian scheme and let $\mathcal{F}$ be an abelian sheaf on the étale site of X . Then the following are equivalent.

- (1) The sheaf $\mathcal{F}$ is a skyscraper sheaf.
- (2) For x closed point of X , denote by $i_x : \text{Spec}(k(x)) \rightarrow X$ the inclusion. Then the units of $(i_x)^* \dashv (i_x)_*$, with $x \in \text{cl}(X)$, furnish an isomorphism $\mathcal{F} \cong \bigoplus_{x \in \text{cl}(X)} (i_x)_*(i_x)^*(\mathcal{F})$.

Proof. The implication (2) $\Rightarrow$ (1) is trivial, since a non-closed point $\bar{x}$ of X does not appear in the sum. The implication (1) $\Rightarrow$ (2) requires more work. First of all, notice that the units of $(i_x)^* \dashv (i_x)_*$, for $x \in \text{cl}(X)$ closed point, induce a morphism

$$\mathcal{F} \rightarrow \prod_{x \in \text{cl}(X)} (i_x)_*(i_x)^*(\mathcal{F}).$$

of abelian sheaves on the étale site of X . We first claim that this map factors through the canonical morphism

$$\bigoplus_{x \in \text{cl}(X)} (i_x)_*(i_x)^*(\mathcal{F}) \rightarrow \prod_{x \in \text{cl}(X)} (i_x)_*(i_x)^*(\mathcal{F})$$

again of abelian sheaves on the étale site of X . If this happens, then, since the étale site has enough points, a computation on stalks allows us to conclude. Indeed, for a closed point x be a closed point of X , the sheaf $(i_x)_*(i_x)^*(\mathcal{F})$ is concentrated in x and the only non trivial stalk, the one at $\bar{x}$, is isomorphic to $\mathcal{F}_{\bar{x}}$.

Since X is quasi-separated (being noetherian), the étale site of X is equivalent to the *restricted étale site*, that is to the site whose category is given by the finitely presented étale schemes over X and surjective families as coverings. In particular, we are left to prove that our favorite morphism factors through the sum when we restrict to finitely presented étale schemes Y over X . In this case it is easy to see that the image of every section in $\mathcal{F}(Y)$ along the morphism $\mathcal{F}(Y) \rightarrow (i_x)_*(i_x)^*(\mathcal{F})(Y)$ is non-zero only for finitely many closed points $x \in \text{cl}(X)$. Here we need X to be noetherian. $\square$

Lemma 2.3. Let X be a noetherian scheme and assume that every closed point has separably closed residue field. Then $H_{\text{ét}}^q(X, \mathcal{F}) = 0$ vanishes for all $q \geq 1$ and skyscraper sheaf $\mathcal{F}$.

Proof. By [Lemma 2.2](#) we have a natural isomorphism

$$\mathcal{F} \cong \bigoplus_{x \in \text{cl}(X)} (i_x)_*(i_x)^*(\mathcal{F}).$$

In particular, we can compute

$$\begin{aligned} H_{\text{ét}}(X, \mathcal{F}) &\cong \bigoplus_{x \in \text{cl}(X)} H_{\text{ét}}(X, (i_x)_*(i_x)^*(\mathcal{F})) \\ &\cong \bigoplus_{x \in \text{cl}(X)} H_{\text{ét}}(\text{Spec}(k(x))_{\text{ét}}, (i_x)^*(\mathcal{F})) \\ &\cong 0. \end{aligned}$$

Let us comment this chain of isomorphisms. In the first one we are exchanging a right adjoint functor with an arbitrary direct sum. This is possible since *pseudo-filtered colimits and global sections commute on a noetherian site*. Let us expand the terminology. Recall first that a noetherian sites is a site where every object

is quasi-compact, meaning that every cover can be refined to a finite cover. Recall also that a category is pseudo-filtered if every cospan can be extended to a commutative square and every pair of parallel arrows can be coequalized (though non universally). Now the étale site of a quasi-separated scheme is equivalent to the restricted étale site. Since the restricted étale site of a quasi-compact scheme is noetherian, the first isomorphism follows since the set of closed points $\text{cl}(X)$ of X , regarded as a discrete category (that is, the homset between two different objects are empty), is a pseudo-filtered category.

Consider now the second isomorphism. Since the residue field $k(x)$ is separably closed, the category of abelian étale sheaves on $\text{Spec}(k(x))$ is equivalent to the category of abelian groups, and the global section functor is exact (since the sheafification is trivial). $\square$

Remark 2.4. The conditions of the previous lemma are clearly satisfied when X is an algebraic scheme over a separably closed field (that is, when X admits a morphism of finite type to the spectrum of a separably closed field). For the purpose of the main theorem, this assumption is enough.

Let X be a noetherian scheme and let us denote by $K(X)$ the field of rational functions on X . Denote also by $j : \text{Spec}(K(X)) \rightarrow X$ the canonical inclusion.

Lemma 2.5. Let X be a noetherian scheme. Let $\mathcal{G} \in \text{Shv}_{\text{Ab}}(\text{Spec}(K(X))_{\text{ét}})$ be an abelian sheaf. Then

$$(\mathbb{R}^q j_*(\mathcal{G}))_{j \circ \bar{\xi}} \cong \begin{cases} \mathcal{G}_{\bar{\xi}} & \text{for } q = 0, \\ 0 & \text{for } q \geq 1, \end{cases}$$

for every maximal point $\xi \in X$ (hence through the finitely many generic points of the irreducible components of X). Here we have regarded $\bar{\xi}$ as a geometric point of $\text{Spec}(K(X))$.

Proof. Recall that for a quasi-compact quasi-separated morphism of schemes $f : Y \rightarrow X$ and for an abelian sheaf $\mathcal{F} \in \text{Shv}_{\text{Ab}}(Y_{\text{ét}})$ the stalk at a geometric point $\bar{x}$ of X of the pushforward can be compute as

$$(\mathbb{R}^q f_*(\mathcal{F}))_{\bar{x}} = H_{\text{ét}}^q(X(\bar{x}) \times_X Y, \bar{\mathcal{F}}),$$

that is, the stalk can be computed as the global sections on the base change of the strict localization in $\bar{x}$ of the inverse image of $\mathcal{F}$. In formulae, the strict localization of X at $\bar{x}$ is the affine scheme $X(\bar{x}) = \text{Spec}(\mathcal{O}_{X, \bar{x}})$ where $\mathcal{O}_{X, \bar{x}}$ is the stalk of the structure sheaf at the geometric point $\bar{x}$.

Now let us apply this observation to our case. We deduce that

$$(\mathbb{R}^q j_*(\mathcal{G}))_{j \circ \bar{\xi}} \cong H_{\text{ét}}^q(X(\bar{\xi}) \times_X \text{Spec}(K(X)), \bar{\mathcal{G}})$$

with obvious meaning of the symbols. Notice that we have confused $X(\bar{\xi})$ with $X(j \circ \bar{\xi})$. We now deduce the claim from the following two facts.

- (1) The base change $X(\bar{\xi}) \times_X \text{Spec}(K(X))$ is isomorphic to $X(\bar{\xi})$.
- (2) Since the field $k(\xi)^{\text{sep}}$ is separably closed, the category of abelian étale sheaves on $\text{Spec}(k(\xi)^{\text{sep}})$ is equivalent to the category of abelian groups, and the global section functor is exact (since the sheafification is trivial).

$\square$

We now use the previous two lemmas to compute the cohomology of arbitrary abelian sheaves on 1-dimensional noetherian schemes.

Proposition 2.6. Let X be a 1-dimensional noetherian scheme and assume that every closed point has separably closed residue field. Let $\mathcal{F} \in \text{Shv}_{\text{Ab}}(X_{\text{ét}})$ be an abelian sheaf on the étale site of X . Then the unit of $j^* \dashv j_*$ induces an equivalence

$$H_{\text{ét}}^q(X, \mathcal{F}) \cong H_{\text{ét}}^q(X, j_* j^*(\mathcal{F}))$$

for every $q \geq 2$.

Proof. Let $\xi \in X$ be a maximal point and regard it as a geometric point of $\text{Spec}(K(X))$. We have then isomorphisms

$$\begin{aligned} \mathcal{F}_{j_{\circ \bar{\xi}}} &\cong (j^* \mathcal{F})_{\bar{\xi}} \\ &\cong (j_* j^*(\mathcal{F}))_{\bar{\xi}}. \end{aligned}$$

Here in the first isomorphism we have used the explicit definition of stalk (and the fact that associating to a morphism its pullback is a 2-functor) whereas in the second we have used [Lemma 2.5](#) applied to $\mathcal{G} = j^*(\mathcal{F})$. Anyway, this computation tells us that the kernel and cokernel of the unit $u : \mathcal{F} \rightarrow j_* j^*(\mathcal{F})$ of $j^* \dashv j_*$ are skyscraper sheaves on the étale site of X . By pasting the long exact sequences in cohomology associated to short exact sequences

$$0 \rightarrow \ker(u) \rightarrow \mathcal{F} \rightarrow \text{im}(u) \rightarrow 0 \quad \quad 0 \rightarrow \text{im}(u) \rightarrow j_* j^*(\mathcal{F}) \rightarrow \text{coker}(u) \rightarrow 0$$

and by using [Lemma 2.3](#) we conclude. $\square$

Proposition 2.7. Let X be a 1-dimensional noetherian scheme and assume that every closed point has separably closed residue field. Let $\mathcal{G} \in \text{Shv}_{\text{Ab}}(\text{Spec}(K(X))_{\text{ét}})$ be an abelian sheaf on the spectrum of the field of rational functions on X . Then we have a long exact sequence

$$0 \rightarrow H_{\text{ét}}^1(X, j_*(\mathcal{G})) \rightarrow H_{\text{ét}}^1(\text{Spec}(K(X)), \mathcal{G}) \rightarrow \bigoplus_{x \in \text{cl}(X)} H_{\text{ét}}^1(X(x) \times_X \text{Spec}(K(X)), \mathcal{G}) \rightarrow H_{\text{ét}}^2(X, j_*(\mathcal{G})) \rightarrow \dots$$

Here we have abused notation. Indeed, the sheaf $\mathcal{G}$ appearing in the third term of this long exact sequences is actually the pullback of $\mathcal{G}$ to the scheme $X(x) \times_X \text{Spec}(K(X))$.

Proof. The proof will be a careful analysis of a certain spectral sequence. First of all, notice that we have an isomorphism $\Gamma_{\text{ét}}(\text{Spec}(K(X)), -) \cong \Gamma_{\text{ét}}(X, j_*(-))$. In particular, we can consider the Leray spectral sequence

$$E_2^{p,q} = H_{\text{ét}}^p(X, \mathbb{R}^q j_*(\mathcal{G})) \Rightarrow H_{\text{ét}}^{p+q}(\text{Spec}(K(X)), \mathcal{G}).$$

Since [Lemma 2.5](#) instructs us that $\mathbb{R}^q j_*(\mathcal{G})$ is a skyscraper sheaf on the étale site of X for every $q \geq 1$, we can apply [Lemma 2.3](#) and deduce that

$$H_{\text{ét}}^p(X, \mathbb{R}^q j_*(\mathcal{G})) = 0$$

for all $q \geq 1$ and $p \geq 1$. It follows that our spectral sequence exists only on a particular region of the plane $\mathbb{Z} \times \mathbb{Z}$. Indeed, originally it lived in the first quadrant, that is in $\mathbb{N} \times \mathbb{N}$, but now it lives only on the coordinate axis! Hence the only non trivial terms are $E_2^{p,0}$ and $E_2^{0,q}$ for $p, q \geq 0$. We deduce the existence of a long exact sequence

$$0 \rightarrow E_2^{1,0} \rightarrow E_{\infty}^1 \rightarrow E_2^{0,1} \rightarrow E_2^{2,0} \rightarrow \dots$$

In particular, the claim will be proved once we have identified the terms $E_2^{p,0}$ and $E_2^{0,q}$ for $p, q \geq 0$. Since

$$E_2^{p,0} = H_{\text{ét}}^p(X, H_{\text{ét}}^0 j_*(\mathcal{G})) = H_{\text{ét}}^p(X, j_*(\mathcal{G}))$$

and since

$$\begin{aligned} E_2^{0,q} &= H_{\text{ét}}^0(X, \mathbb{R}^q j_*(\mathcal{G})) \\ &\cong H_{\text{ét}}^0(X, \bigoplus_{x \in \text{cl}(X)} (i_x)_* (i_x)^* \mathbb{R}^q j_*(\mathcal{G})) \\ &\cong \bigoplus_{x \in \text{cl}(X)} H_{\text{ét}}^0(X, (i_x)_* (i_x)^* \mathbb{R}^q j_*(\mathcal{G})) \\ &\cong \bigoplus_{x \in \text{cl}(X)} H_{\text{ét}}^0(\text{Spec}(k(x)), (i_x)^* \mathbb{R}^q j_*(\mathcal{G})) \\ &\cong \bigoplus_{x \in \text{cl}(X)} (\mathbb{R}^q j_*(\mathcal{G}))_{\bar{x}} \\ &\cong \bigoplus_{x \in \text{cl}(X)} H_{\text{ét}}^q(X(\bar{x}) \times_X \text{Spec}(K(X)), \mathcal{G}) \end{aligned}$$

we are done. Let us comment these isomorphisms for the sake of clarity. The first one follows since $\mathbb{R}^q j_*(\mathcal{G})$ is a skyscraper sheaf on the étale site of X for every $q \geq 1$. The second isomorphism follows instead since $X_{\text{ét}}$ is a noetherian site, being X quasi-compact and quasi-separated. The third isomorphism follows by adjunction, and the fourth one follows by definition of stalk. The fifth one follows instead by the explicit computation of the stalk of the pushforward. $\square$

3 Cohomological dimension of algebraic curves

In order to introduce the cohomological dimension of quasi-compact quasi-separated schemes we need to introduce the notion of l -torsion sheaf for a prime l . A simple as it might be, an l -torsion sheaf is not a sheaf of abelian groups which takes values in the full subcategory of l -torsion abelian groups. Instead, an abelian sheaf $\mathcal{F} \in \text{Shv}_{\text{Ab}}(\mathcal{C})$ on a Grothendieck site $(\mathcal{C}, \tau)$ is an l -torsion sheaf if it is the sheafification of a presheaf of abelian torsion groups.

Remark 3.1. Let $(\mathcal{C}, \tau)$ and let $\mathcal{F} \in \text{Shv}_{\text{Ab}}(\mathcal{C})$ be an abelian sheaf. There is a subtle difference between $\mathcal{F}$ being an l -torsion sheaf and a $\mathcal{F}$ taking values in the full subcategory of l -torsion abelian groups. This difference disappears if we restrict to quasi-compact objects of our site $(\mathcal{C}, \tau)$. Clearly, on noetherian sites this difference disappears since every object is quasi-compact. In particular, this happens if $(\mathcal{C}, \tau)$ is the étale site of a quasi-compact quasi-separated scheme.

Definition 3.2. Let X be a quasi-compact quasi-separated scheme and l be a prime number. The *cohomological l -dimension* of X is the smallest (extended) integer $\text{cd}(X) \in \mathbb{N} \cup \{\infty\}$ such that:

- (1) The étale cohomology $H_{\text{ét}}^q(X, \mathcal{F}) = 0$ vanishes for every $q \geq n + 1$ and for every l -torsion sheaf $\mathcal{F}$.
- (2) The l -primary component of the étale cohomology $H_{\text{ét}}^q(X, \mathcal{F})(l) = 0$ vanishes for every $q > n$ and for every l -torsion sheaf $\mathcal{F}$. Recall that the l -primary component of an abelian group is defined to be the subgroup of all elements whose order is a power of l .

We will denote by $\text{cd}(X) = \max_l \text{cd}_l(X)$ the *cohomological dimension* of X .

Lemma 3.3. Let X be a 1-dimensional noetherian scheme and assume that every closed point has separably closed residue field. Let l be a prime and assume that $\mathrm{cd}_l(K(X)) \leq r$. Consider also an l -torsion sheaf $\mathcal{G} \in \mathrm{Shv}_{\mathrm{Ab}}(\mathrm{Spec}(K(X))_{\mathrm{\acute{e}t}})$. Then:

- (1) The higher cohomology groups $H_{\mathrm{\acute{e}t}}^q(X, j_*\mathcal{G}) = 0$ vanish for $q \geq r + 1$.
- (2) There exists a long exact sequence

$$\begin{aligned} 0 \rightarrow H_{\mathrm{\acute{e}t}}^1(X, j_*(\mathcal{G})) &\rightarrow H_{\mathrm{\acute{e}t}}^1(\mathrm{Spec}(K(X)), \mathcal{G}) \rightarrow \bigoplus_{x \in \mathrm{cl}(X)} H_{\mathrm{\acute{e}t}}^1(X(x) \times_X \mathrm{Spec}(K(X)), \mathcal{G}) \rightarrow \\ &\rightarrow H_{\mathrm{\acute{e}t}}^2(X, j_*(\mathcal{G})) \rightarrow \cdots \rightarrow \\ &\rightarrow H_{\mathrm{\acute{e}t}}^{r+1}(X, j_*(\mathcal{G})) \rightarrow 0. \end{aligned}$$

Here, as before, we have abused notation since the sheaf $\mathcal{G}$ appearing in the third term of this long exact sequences is actually the pullback of $\mathcal{G}$ to the scheme $X(x) \times_X \mathrm{Spec}(K(X))$.

Proof. Consider first point (2). Since the cohomological dimension $\mathrm{cd}_l(K(X)) \leq r$ is bounded by r , the term $H_{\mathrm{\acute{e}t}}^q(\mathrm{Spec}(K(X)), \mathcal{G}) = 0$ vanishes for every $q \geq r + 1$. In particular, the long exact sequence of [Proposition 2.7](#) allows us to conclude.

Point (1) is a bit more complicated and will sue the only black box of these pages: we claim that, trough Galois cohomology, one can show that for a noetherian scheme X , whose residue field at closed point are not necessarily spearily closed, we have

$$\mathrm{cd}_l(X(\bar{x}) \times_X \mathrm{Spec}(K(X))) \leq \mathrm{cd}_l(K(X))$$

for every geometric point $\bar{x}$ of X . With this fact in our hands, the long exact sequence of [Proposition 2.7](#) allows us to deduce that

$$H_{\mathrm{\acute{e}t}}^q(X, j_*(\mathcal{G})) \cong \bigoplus_{x \in \mathrm{cl}(X)} H_{\mathrm{\acute{e}t}}^{q-1}(X(x) \times_X \mathrm{Spec}(K(X)), \mathcal{G})$$

is zero when $q > r + 1$. Here we have used that the pushfoward j_* preserves l -torsion sheaves since j is a map between noetherian schemes. $\square$

Theorem 3.4. Let X be a 1-dimensional noetherian scheme and assume that every closed point has separably closed residue field. Let l be a prime. Then $\mathrm{cd}_l(X) \leq \mathrm{cd}_l(K(X)) + 1$, and so $\mathrm{cd}(X) \leq \mathrm{cd}(K(X)) + 1$.

Proof. If $\mathrm{cd}_l(K(X)) = \infty$ then there is nothing to prove. Assume therefore that the cohomological l -dimension of the field of rational functions $K(X)$ on X is finite. Pick in particular an integer q such that $q > \mathrm{cd}_l(K(X)) + 1$. Let $\mathcal{F} \in \mathrm{Shv}_{\mathrm{Ab}}(X_{\mathrm{\acute{e}t}})$ be an l -torsion sheaf. We wish to show that the q -th étale cohomology group $H_{\mathrm{\acute{e}t}}^q(X, \mathcal{F}) = 0$ vanishes. This is immediate: since $q \geq 2$ we can apply [Proposition 2.6](#) to deduce

$$H_{\mathrm{\acute{e}t}}^q(X, \mathcal{F}) \cong H_{\mathrm{\acute{e}t}}^q(X, j_*j^*(\mathcal{F}))$$

so that point (1) of [Lemma 3.3](#) concludes. $\square$

4 The proof

We are finally ready to put everything together.

Proposition 4.1. Let X be a 1-dimensional noetherian scheme and assume that every closed point has separably closed residue field. Let n be a natural number invertible on X and assume that $\text{cd}_l(K(X)) \leq 1$ for all primes l dividing n . Then:

- (1) For $q > 2$ we have $H_{\text{ét}}^q(X, (\mu_n)_X) = 0$.
- (2) For $q \leq 2$ the cohomology of $(\mu_n)_X$ is given by the exact sequence

$$\begin{aligned} 0 \rightarrow H_{\text{ét}}^0(X, (\mu_n)_X) &\rightarrow \Gamma(X, \mathcal{O}_X^*) \xrightarrow{n} \Gamma(X, \mathcal{O}_X^*) \\ &\rightarrow H_{\text{ét}}^1(X, (\mu_n)_X) \rightarrow \text{Pic}(X) \xrightarrow{n} \text{Pic}(X) \\ &\rightarrow H_{\text{ét}}^2(X, (\mu_n)_X) \rightarrow 0. \end{aligned}$$

of abelian groups.

Proof. We prove point (1) by considering two cases: when the prime l divides n and when it doesn't. In the first case we can use [Theorem 3.4](#), which actually holds for all primes l , to deduce that $\text{cd}_l(X) \leq \text{cd}(K(X)) + 1$. Since $\text{cd}_l(K(X)) \leq 1$ holds when l divides n , we deduce that $\text{cd}_l(X) \leq 2$. Now, by considering the sheaf of n -th roots of unity $(\mu_n)_X$ and by taking its l -component pointwise, we construct an l -torsion sheaf $(\mu_n)_X(l)$ on the étale site of X . By definition of cohomological dimension we deduce that

$$H_{\text{ét}}^q(X, (\mu_n)_X(l)) = 0$$

for every $q > 2$. **If l and n are coprimes, we may assume that n is a prime l' . In this case $(\mu_n)_X(l') = 0$, so that, a fortiori**

$$H_{\text{ét}}^q(X, (\mu_n)_X)(l') = H_{\text{ét}}^q(X, (\mu_n)_X(l')) = 0$$

for $q > 2$ (because it holds for every $q \geq 0$). Anyway, these two observations tell us that all the possible components are always zero, we deduce that $H_{\text{ét}}^q(X, (\mu_n)_X) = 0$ for $q \geq 0$.

Consider now point (2). Since n is invertible on X , the Kummer sequence

$$0 \rightarrow (\mu_n)_X \rightarrow (\mathbb{G}_m)_X \xrightarrow{n} (\mathbb{G}_m)_X \rightarrow 0$$

is exact. By taking the associated long exact sequence in cohomology we deduce the long exact sequence we have claimed. Indeed, since we have isomorphisms

$$H_{\text{ét}}^0(X, (\mathbb{G}_m)_X) \cong \Gamma(X, \mathcal{O}_X^*), \quad H_{\text{ét}}^1(X, (\mathbb{G}_m)_X) \cong \text{Pic}(X)$$

the first one by definition of derived functor and the second one by the infamous Hilbert's Theorem 90, the claim will be proved once we have observed that $H_{\text{ét}}^2(X, (\mathbb{G}_m)_X) \cong 0$. Here is the second big black box of these pages: one can show that the l -primary component of cohomology group $H_{\text{ét}}^2(X, (\mathbb{G}_m)_X)(l) = 0$ vanishes for all primes l dividing n . This implies that the multiplication by n induces an injective map $H_{\text{ét}}^2(X, (\mathbb{G}_m)_X) \rightarrow H_{\text{ét}}^2(X, (\mathbb{G}_m)_X)$. $\square$

Remark 4.2. Let us comment on the black box we have just used. Let X be a 1-dimensional noetherian scheme and assume that every closed point has separably closed residue field. The vanishing of $H_{\text{ét}}^2(X, (\mathbb{G}_m)_X)(l) = 0$ for all primes l invertible on X with $\text{cd}_l(X) \leq 1$ can be justified as follows.

By pulling back $(\mathbb{G}_m)_X$ along $j : \text{Spec}(K(X)) \rightarrow X$ we realize that $j^*((\mathbb{G}_m)_X)$ gets identified with $(\mathbb{G}_m)_{\text{Spec}(K(X))}$. Therefore we have to show that the l -primary component of $H_{\text{ét}}^q(X, j_*(\mathbb{G}_m)_{\text{Spec}(K(X))})$ vanishes for $q \geq 2$. The exact sequence of [Proposition 2.7](#), which becomes interesting for $q \geq 1$, allows us to conclude: by means of Galois cohomology and the Kummer sequence associated to l (which is invertible on $\text{Spec}(K(X))$) we deduce that the l -primary components vanish.

The main theorem of these notes is then a consequence of the previous result. Let us recall its statement.

Theorem 4.3. Let k be a separably closed field and let X/k be a connected complete algebraic curve. Let n be an integer relative prime to the characteristic of k . Then the étale cohomology of the sheaf of primitive n -th roots of unit on X can be computed as

$$H_{\text{ét}}^*(X, (\mu_n)_X) = \begin{cases} \mu_n(X) & \text{for } * = 0, \\ {}_n\text{Pic}(X) & \text{for } * = 1, \\ \text{Pic}_n(X) & \text{for } * = 2, \end{cases}$$

for lower degrees. In higher degrees it vanishes.

Proof. First of all, let us argue why for a X/k over a separably closed field the l -cohomological dimension $\text{cd}_l(K(X)) \leq 1$ is bounded by 1. Since $\text{cd}_l(K(X)) = \max_{\xi} \text{cd}_l(k(\xi))$ can be computed as the maximum of the l -cohomological dimensions of the residue fields at the maximal points of X , it suffices to show that they are all bounded by 1. However, since X is an algebraic curve, they have transcendence degree 1 over the separably closed field k . But now Tsen's theorem implies that $\text{cd}_l(k(\xi)) \leq 1$.

We now apply the previous proposition: by using the fact that multiplication by n is surjective on global sections (being relative prime to the characteristic of k it is invertible on X , we conclude. $\square$